

SALES ANALYTICS

Internship Task 2 — Professional Project Report

Sales Analysis using Python, SQL, PostgreSQL & Power BI

Internship	Sumerix Global
Task	Task 2 — Sales Analytics
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Program	B.E. / B.Tech Computer Engineering

Prepared for internship submission

1. Executive Summary

This project presents an end-to-end sales analytics workflow for Internship Task 2. The work covers data preparation, PostgreSQL/SQL analysis, Python/Jupyter analysis, and Power BI visualization.

The final analysis contains 5,000 orders and reports approximately ₹12.51 million in total sales, ₹2.24 million in total profit, 27,507 units sold, and an average order value of approximately ₹2,501. The Power BI dashboard summarizes category, regional, segment, time, customer, and product performance.

2. Project Objectives

- Analyze overall sales, profit, orders, and quantity performance.
- Compare performance across categories, regions, and customer segments.
- Identify high-performing customers and products.
- Study monthly, quarterly, and yearly sales behavior.
- Use SQL and Python for structured analysis and exploratory analysis.
- Build a Power BI dashboard for business-oriented visualization.

3. Dataset Overview

The cleaned sales dataset contains transactional sales information covering orders, customers, products, geography, financial measures, and time-related fields.

Field	Purpose
Order ID	Unique order identifier
Order Date / Ship Date	Order and shipping dates
Ship Mode	Shipping method
Customer ID / Customer Name	Customer identification
Segment	Customer segment
Region	Geographic region

Category / Sub-Category	Product classification
Product Name	Product-level analysis
Sales	Sales value
Quantity	Units sold
Discount	Discount applied
Profit	Profit generated
Order Year / Month / Quarter	Time-based analysis
Shipping Days	Derived shipping duration

4. Data Cleaning & Preparation

- Reviewed the dataset structure and prepared fields for analysis.
- Verified and organized date-related fields for year, month, quarter, and weekday analysis.
- Used derived fields including Order Year, Order Month, Order Quarter, and Shipping Days.
- Prepared the cleaned data for PostgreSQL, Jupyter/Pandas, and Power BI.

5. SQL & PostgreSQL Analysis

PostgreSQL was used as the relational database layer. SQL queries were used to aggregate, classify, compare, and analyze the sales data.

- SUM, COUNT, AVG and grouped aggregation.
- GROUP BY analysis for category, region, segment, customer, and time periods.
- CASE WHEN logic for sales categorization and profit-status classification.
- Customer and regional analysis.
- Yearly sales growth and quarterly analysis.
- Profitability and margin analysis.
- Analytical/window-function work where required by the task.

6. Python / Jupyter Analysis

Python was used for exploratory and analytical work with Pandas and NumPy. Jupyter provided a reproducible environment for calculations and validation.

- Loaded and inspected the cleaned dataset.
- Performed grouped and time-based aggregations.
- Calculated monthly sales and business KPIs.
- Analyzed customer, segment, category, region, and quarterly performance.
- Calculated yearly sales growth.

7. Key Performance Indicators

KPI	Result
Total Sales	₹12.51M
Total Profit	₹2.24M
Total Orders	5,000
Total Quantity	27,507

Average Order Value	≈ ₹2,501.36
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8. Analytical Results

8.1 Segment Analysis

Segment	Sales	Profit
Corporate	₹4,244,266.85	₹732,570.05
Home Office	₹4,136,313.47	₹775,561.84
Consumer	₹4,126,235.19	₹731,579.24

8.2 Regional Analysis

Region	Sales	Profit
South	₹3,247,440.79	₹574,301.38
West	₹3,175,737.09	₹569,469.81
Central	₹3,084,820.38	₹558,202.61
Region	Sales	Profit
East	₹2,998,817.25	₹537,737.33

8.3 Category Profitability

Category	Sales	Profit	Margin
Furniture	₹4,169,553.39	₹753,823.28	18.08%
Office Supplies	₹4,184,882.48	₹754,427.39	18.03%
Technology	₹4,152,379.64	₹731,460.46	17.62%

8.4 Yearly Sales Performance

Year	Sales	YoY Growth
2023	₹4,600,381.57	—
2024	₹6,171,980.63	34.16%
2025	₹1,734,453.31	-71.90%

8.5 Quarterly Analysis

Quarter	Sales	Profit	Margin
Q1	₹3,004,572.67	₹524,372.55	17.45%
Q2	₹3,185,401.16	₹574,273.43	18.03%
Q3	₹3,157,588.84	₹560,533.18	17.75%
Q4	₹3,159,252.84	₹580,531.97	18.38%

Note: These yearly and quarterly figures are the values produced during the project analysis; no cause is inferred for observed changes.

9. Power BI Dashboard

The final Power BI dashboard consolidates the analytical results into a business-facing view. It contains KPI cards and visual analysis for category, region, monthly trend, customer segment, quarterly performance, top products, and top customers.

- Total Sales, Total Profit, Total Orders, Total Quantity, and Average Order Value.

- Sales by Category and Sales by Region.
- Monthly Sales Trend and Profit by Category.
- Sales by Customer Segment and Profit by Region.
- Quarterly Sales & Profit.
- Top 10 Products by Profit and Top 10 Customers by Sales.

10. Key Insights

- Total sales are approximately ₹12.51M and total profit is approximately ₹2.24M.
- The dataset contains 5,000 orders and 27,507 units sold.
- Average order value is approximately ₹2,501.36.
- Sales are distributed across Corporate, Home Office, and Consumer segments with relatively close totals.
- Regional analysis compares South, West, Central, and East across sales and profit.
- The three main categories have broadly similar sales totals and profit margins around 17.6%–18.1%.
- Yearly analysis shows higher sales in 2024 than 2023, followed by a lower 2025 figure in this dataset.
- Quarterly sales are relatively close across Q1–Q4, while profit and margin vary.

11. Tools & Technologies

- Python, Pandas, NumPy, and Jupyter Notebook.
- PostgreSQL and SQL.
- Power BI for dashboarding and business visualization.

12. Conclusion

This project demonstrates an end-to-end sales analytics workflow from data preparation and SQL analysis to Python exploration and Power BI visualization. The resulting dashboard provides a concise view of sales, profit, customer, product, regional, category, and time-based performance and creates a foundation for further statistical, forecasting, or machine-learning work.

13. Project Deliverables

- Cleaned sales dataset
- PostgreSQL database/table
- SQL analysis queries
- Jupyter Notebook
- Power BI (.pbix) dashboard
- Dashboard PDF/export
- Professional project report